PhishShift: Message Risk Scanner

Project Evaluation Report

Purpose

PhishShift is an educational message-risk screening tool. It classifies SMS-like text and highlights warning cues. It is not a definitive phishing detector and does not open submitted URLs.

Evaluation scope

The metrics below are from the project's recorded random, stratified ham/spam test split. Spam labels are not equivalent to verified phishing labels. Campaign-separated evaluation remains incomplete because message-to-campaign mapping is not reliable.

Recorded model metrics

Model	Accuracy	Precision	Recall	F1	FPR	TN	FP	FN	TP
Rules baseline	89.53%	91.67%	17.19%	28.95%	0.22%	902	2	106	22
Multinomial Naive Bayes	96.32%	100.00%	70.31%	82.57%	0.00%	904	0	38	90
Logistic Regression	97.87%	90.77%	92.19%	91.47%	1.33%	892	12	10	118

Highest recorded F1: Logistic Regression (F1 91.47%, accuracy 97.87%). These scores should not be interpreted as phishing-specific performance.

Known limitations and outstanding work

Near-duplicate removal is not verified. Campaign-separated testing is not complete. The safe abstain/manual-review behavior must be implemented and tested before being marked complete. NCSU messages with unknown labels are not used as supervised training labels. Dataset-specific reuse terms should be confirmed before redistribution.

Safety guidance

Never open a suspicious link to test it. Verify unexpected requests using an independently obtained official app, website, or phone number. Do not enter passwords, PINs, OTPs, or banking information into untrusted pages.

Evidence files

Recorded metrics: results/model_metrics.csv; confusion matrices: results/confusion_matrices.json; evaluation summary: results/evaluation_summary.json; requirement tracker: docs/REQUIREMENT_TRACKER.md.